

ENVIRONMENTAL NEGOTIATORS

Student Setup Guide — Windows & macOS

Work through each step top-to-bottom. Where the command differs between operating systems, you'll see two subheadings — **On Windows** and **On macOS**. Follow the one for your machine and ignore the other.

Grey boxes are commands to type into the terminal. Yellow boxes flag things that commonly trip people up.

Step 1 — Install Python

Already have this? Run `python --version` (Windows) or `python3 --version` (macOS). If it shows 3.10 or higher, skip to Step 2.

Python is the language the workshop code is written in. You need version 3.10 or newer.

Go to python.org/downloads and download the latest Python 3 installer for your OS. Run it.

Windows — tick 'Add python.exe to PATH' on the first installer screen.

If you miss this checkbox, every command in this guide will fail. If you already installed without it, uninstall and reinstall with the box checked.

macOS — always use python3, not python.

Plain 'python' often points at an old built-in version or doesn't exist. Every Python command in this guide uses python3 on Mac for that reason.

Verify the install:

On Windows

```
python --version
```

On macOS

```
python3 --version
```

You should see Python 3.10 or higher. If you see 'command not found' or 'not recognized', close the terminal, open a new one, and try again.

Step 2 — Install VS Code

Already have this? If VS Code (or another proper code editor like PyCharm or Sublime) is already installed, skip to Step 3.

VS Code is the editor you'll use to open and edit the workshop code.

Go to **code.visualstudio.com**, download the installer for your OS, run it, and accept the defaults.

Do NOT use Notepad, TextEdit, or Word to edit code.

They silently mangle indentation and your code will break in ways that are painful to debug. Any real code editor is fine — just not those three.

Step 3 — Download and unzip the workshop folder

Download the workshop .zip file your instructor sent you. Unzip it to a location you'll remember — Desktop or Downloads is fine.

After unzipping, you'll have a folder named **environmental-negotiators-student**. Open it — inside is a subfolder called **negotiators**. That inner negotiators folder is where everything happens.

Step 4 — Open a terminal INSIDE the negotiators folder

This is the step students mess up most — read carefully.

Every command in the rest of this guide only works if your terminal is currently sitting inside the negotiators folder. If you open a terminal from the Start menu or Spotlight, it starts in your home folder — that's wrong. Use one of the methods below instead.

On Windows

Easiest method — right-click in File Explorer:

- Open **File Explorer** and navigate into **environmental-negotiators-student** until you can see the **negotiators** folder.
- **Right-click** the **negotiators** folder.
- Windows 11: click **Open in Terminal**.
- Windows 10: hold **Shift** while right-clicking, then choose **Open PowerShell window here**.

Alternative — type the path manually in Command Prompt:

```
cd "C:\path\to\environmental-negotiators-student\negotiators"
```

Confirm you're in the right place:

```
dir
```

You should see `negotiators.py`, `requirements.txt`, `verify.py`, and a backend folder. If you don't, you're in the wrong place — try again.

On macOS

Easiest method — drag the folder into Terminal:

- Open **Terminal** (Cmd+Space, type 'Terminal', hit Enter).
- Type `cd` in the terminal (*note the space after cd — don't hit Enter yet*).
- Open **Finder**, navigate into `environmental-negotiators-student`, and find the `negotiators` subfolder.
- **Drag the negotiators folder** from Finder directly onto the Terminal window. macOS pastes its full path after your '`cd`'. Hit Enter.

Alternative — right-click in Finder → Services → New Terminal at Folder.

Confirm you're in the right place:

```
ls
```

You should see `negotiators.py`, `requirements.txt`, `verify.py`, and a backend folder. If you don't, you're in the wrong place — try again.

Keep this terminal open for Steps 5 through 10.

If you close it and open a new one later, you'll need to navigate back into the `negotiators` folder first — same method as above.

Step 5 — Install Ollama

Ollama is the tool that runs the AI model on your computer. Think of it as a translator between the Python code and the AI brain.

Go to ollama.com/download, download the installer for your OS, and run it.

On Windows

After installing, look at your system tray (bottom-right of the screen) for a small llama icon. If you don't see it, open the Ollama app from the Start menu.

On macOS

After installing, look at the top menu bar (top-right of the screen) for a small llama icon. If you don't see it, open the Ollama app from your Applications folder.

If you already had Ollama installed from before — update it.

Open the Ollama app and let it update itself. Older versions may behave unexpectedly with the llama3.1 model.

Step 6 — Download llama3.1 and point the workshop code at it

Three parts: download the model, test it runs, then edit one line of code so the workshop knows which model to use.

6a — Download the model

In your terminal (still inside the negotiators folder), run this. The command is the same on Windows and macOS:

```
ollama pull llama3.1
```

This downloads about 4.7 GB.

On a decent connection it takes a few minutes. On university Wi-Fi it can be slow or get blocked entirely. If it hangs or fails, try your phone hotspot.

If you see 'ollama is not recognized' (Windows) or 'command not found: ollama' (macOS)...

Close the terminal, open a new one, navigate back into the negotiators folder (Step 4 method), and try again. New installs don't show up in terminals that were already open.

6b — Test that it runs

Once the download finishes, start the model to confirm it's working:

```
ollama run llama3.1
```

You'll see a prompt like '>>> Send a message'. That means it's working. Type this and hit Enter to exit back to your normal terminal:

```
/bye
```

Ollama keeps the model available in the background for the workshop.

6c — Edit backend/runner.py so the code uses llama3.1

By default the workshop code is configured for a different model. You need to change one line so it points at the llama3.1 model you just downloaded.

- Open **VS Code**.
- Go to **File** → **Open Folder...** and choose the negotiators folder (the same one your terminal is in).

- In the sidebar on the left, expand the backend folder and click on `runner.py` to open it.
- Scroll to **line 9**. Line numbers appear on the left edge of the editor.

Line 9 currently looks like this:

```
MODEL = os.environ.get("OLLAMA_MODEL", "glm-5.1:cloud")
```

Change the model name so line 9 reads exactly:

```
MODEL = os.environ.get("OLLAMA_MODEL", "llama3.1")
```

Save the file: **Ctrl+S** on Windows, **Cmd+S** on macOS.

If llama3.1 is too slow on your laptop, you can swap to a lighter model.

mistral is a common fallback (smaller, faster, less RAM). To swap: run 'ollama pull mistral' in your terminal to download it, then come back to line 9 of backend/runner.py and change "llama3.1" to "mistral", save the file. For today's workshop, start with llama3.1 — only swap if you hit real performance problems during class.

Step 7 — Install the Python packages

Make sure your terminal is still inside the `negotiators` folder (Step 4). Then run:

On Windows

```
python -m pip install -r requirements.txt
```

On macOS

```
pip3 install -r requirements.txt
```

This installs FastAPI, Uvicorn, the Ollama client, and a few others. It takes 30 seconds to a minute. You'll see a lot of 'Successfully installed...' lines at the end. A few yellow warnings are normal.

Windows — if you see 'Fatal error in launcher'...

Your pip shortcut is broken. Use the command exactly as written above (`python -m pip ...`). If even that fails, try: `py -m pip install -r requirements.txt`

macOS — do NOT use sudo.

If you get a permission error, try the command again without `sudo`. If it still fails, your Python was installed as root — reinstall from python.org.

If you see 'No such file or directory: requirements.txt'...

Your terminal isn't inside the negotiators folder. Go back to Step 4 and fix that first.

Step 8 — Install Node.js

Already have this? Run `node --version`. If it shows v18 or higher, skip to Step 9.

Node.js runs the web dashboard. Check first:

```
node --version
```

If you need to install it, go to **nodejs.org** and download the **LTS** version on the left side of the page (not 'Current'). LTS means long-term support — more stable.

After installing Node, close your terminal and open a new one.

The old terminal doesn't know Node exists yet. Re-do Step 4 to get back into the negotiators folder, then keep going.

Step 9 — Install the dashboard packages

From inside the negotiators folder, run these three commands one at a time. The command is the same on Windows and macOS:

```
cd frontend  
npm install  
cd ..
```

This takes one to two minutes. You'll see a lot of warnings — **ignore them**. The `cd ..` at the end puts you back in the main negotiators folder, which matters for the next step.

If you see 'npm is not recognized' / 'command not found: npm'...

You either skipped Step 8 or didn't restart your terminal after installing Node. Close the terminal, open a new one, navigate back into the negotiators folder, and try again.

Step 10 — Verify everything works

Run the verification script — it checks everything you just did in one go:

On Windows

```
python verify.py
```

On macOS

```
python3 verify.py
```

You should see 9 or 10 green plus-signs. The one check that's allowed to fail is 'Backend reachable' — that's only relevant once the backend server is running (Step 11).

Expected output:

```
Environmental Negotiators – Workshop Verification
```

```
[+] Python 3.11.0
[+] Python package: fastapi
[+] Python package: uvicorn
[+] Python package: ollama
[+] Ollama running
[+] Model: llama3.1
[+] Node.js 20.0.0
[+] Frontend dependencies
[+] Agent class
[X] Backend reachable – Start with: ...
```

```
Results: 9/10 checks passed
```

If you see an [X] on anything other than 'Backend reachable', check the troubleshooting table at the end of this guide.

Step 11 — Run the agents

Installation done. Now you'll start the two servers that make the dashboard work. You need TWO terminal windows open at the same time — one for the backend (the AI brain), one for the frontend (the dashboard web page).

Before you start — confirm Ollama is running.

Look for the llama icon (Windows system tray / Mac top menu bar). If you don't see it, open the Ollama app first. The backend will crash on startup if Ollama isn't running.

Terminal #1 — start the backend

Open your first terminal and make sure it's inside the negotiators folder (Step 4 method). Then run:

On Windows

```
python -m uvicorn backend.main:app --reload --port 8002
```

On macOS

```
python3 -m uvicorn backend.main:app --reload --port 8002
```

You should see log lines ending with:

```
INFO:      Uvicorn running on http://127.0.0.1:8002
INFO:      Application startup complete.
```

LEAVE this terminal running the whole time you're using the workshop.

Do not close it. If you close it, the dashboard dies. If you need to stop the server later, press Ctrl+C in this terminal.

Terminal #2 — start the frontend

Open a SECOND terminal (don't close Terminal #1) and navigate it into the `negotiators` folder too — same Step 4 method. Then run these two commands one at a time. The commands are the same on Windows and macOS:

```
cd frontend
npm run dev
```

You should see:

```
VITE v5.x.x  ready in 500 ms
→ Local:   http://localhost:5173/
```

LEAVE this terminal running too.

Both servers — backend and frontend — must stay up while you use the dashboard.

Open the dashboard

In Chrome or Edge, go to: **`http://localhost:5173`**

Check the header — it should say **'2 agents online'** (the Site Researcher and the Manager — you'll add the advocates during the workshop).

If the header says '0 agents online'...

Look at Terminal #1 — there's probably red error text meaning `negotiators.py` has a syntax error. Your instructor will walk you through fixing this during class.

When class is over and you want to shut everything down, go to each terminal and press Ctrl+C. Close the browser tab. Done.

Troubleshooting

Scan the middle column for what you're seeing. The fix is in the right column. The OS column tells you which system the row applies to (Both = applies to both).

OS	What you see	What to do
Windows	'python' is not recognized	You installed Python without 'Add python.exe to PATH'. Reinstall with that box checked, or use 'py' instead of 'python' everywhere.
Windows	'pip' is not recognized	Same root cause. Use 'python -m pip' or 'py -m pip' everywhere this guide says 'pip'.
Windows	'Fatal error in launcher'	The pip shortcut is broken. Use 'python -m pip install -r requirements.txt' instead of plain 'pip'.
macOS	'python' not found	On Mac use 'python3', not 'python'. Same for pip — use 'pip3'. This applies to every command in this guide.
macOS	Permission error ('EACCES')	Don't use sudo. Try the command again without it. If it still fails, your Python was installed as root — reinstall from python.org.
Both	'ollama' command not found	Close the terminal, open a new one, navigate back into the negotiators folder (Step 4), and try again. New installs need a fresh terminal.
Both	'No such file: requirements.txt'	Your terminal isn't inside the negotiators folder. Go back to Step 4.
Both	'ollama pull' hangs forever	Your network is probably blocking Ollama. Switch to a phone hotspot and try again.
Both	npm install fails or hangs	Press Ctrl+C to cancel. Delete the 'frontend/node_modules' folder if it was partially created. Make sure you ran 'cd frontend' first. Try again.
Both	verify.py says model not found	You skipped 'ollama pull llama3.1' in Step 6, or it failed silently. Run it again.
Both	verify.py says Ollama not running	The Ollama app isn't open. Start it — look for the llama icon in the system tray (Windows) or top menu bar (Mac).
Both	verify.py: only 'Backend reachable' fails	That's expected until you reach Step 11. Your setup is fine.

OS	What you see	What to do
Both	Browser shows 'Connection refused'	Terminal #1 (backend) isn't running. Start it again — see Step 11.
Both	Browser page is blank	Terminal #2 (frontend) isn't running. Inside the frontend folder, run 'npm run dev' again.
Both	Header says '0 agents online'	Syntax error in negotiators.py. Check Terminal #1 for red text. Usually a missing comma between Agent() blocks or an unclosed quote. Fix, save, and the header will update within seconds.